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Digital twin enabled cellular network management and prediction

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Abstract

Digital twin (DT) technologies have been increasingly important and useful for wireless communications. In particular, to support soaring wireless traffic with limited frequency spectrum assigned to legacy cellular systems, efficient operation and management of wireless resources as well as preemptively prediction of future spectrum usage are crucially important. For such purpose, DT networks for current fourth-generation (4G) and fifth-generation (5G) networks are constructed in this paper, by simultaneously utilizing measurement data from user equipments (UEs) and geographical information and characteristics of 4G and 5G base stations (BSs) within a specific observation area. Representative case studies are provided to demonstrate the usefulness of DT enabled cellular network management and prediction. As a real or near real time DT application, the impact and benefit of dual connectivity and dynamic spectrum sharing between 4G and 5G networks are analyzed. As a non-real time DT application, long-term improvement of 5G networks such as densification of BSs, implementation of advanced multiple input and multiple output technologies, and assignment of additional spectrum are analyzed and compared.
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Keywords: 5G; Digital twin; Dual connectivity; Dynamic spectrum sharing; Advanced MIMO

1. Introduction

The mobile communication industry has experienced substantial growth over generations and such growth is expected to accelerate in the upcoming sixth-generation (6G) cellular systems [1]. It has been reported in [2] that fifth-generation (5G) mobile subscriptions globally exceeded 1 billion in 2022 and are expected to reach 4.4 billion by 2027, accounting for 48% of all mobile subscriptions. As a consequence, the price of legacy spectrum bands has experienced a sharp rise in recent years due to soaring traffic demand [3]. As reported in [4], 5G mobile network operators (MNOs) in South Korea spent a total of $3.3 billion for operating two bands on 3.5 GHz and 28 GHz.

To secure a stable frequency spectrum, the Federal Communications Commission (FCC) stipulated that the 28 GHz spectrum band is used for 5G system development. The 3rd Generation Partnership Project (3GPP) divided the 5G band into the frequency range 1 (FR1) (sub-6 GHz frequency, 410 MHz– 7125 MHz) and the FR2 (millimeter wave frequency, 24.25 GHz–52.6 GHz) in Release 15 [5]. However, due to several technical and practical limitations of ultra-high frequency communication, efficient and flexible utilization of the currently operating fourth-generation (4G) and 5G spectrum bands, i.e., sub-6 GHz, might be crucially important along with the development of the upper-mid band for 6G, i.e., 7 to 24 GHz, to provide enhanced throughput and cell coverage [6,7]. There exist several methodologies to improve the spectral efficiency (SE) or area capacity of 4G or 5G systems and to support flexible utilization of 4G and 5G spectrum resources. One of the straightforward methods for improving area capacity is to equip additional base stations (BSs) in congested areas, for instance, hot spots or small cells in dense urban environments. Another method is to replace or upgrade BSs with more advanced ones. Moreover, dual connectivity (DC) between 4G and 5G non-standalone (NSA) systems, as well as dynamic spectrum sharing (DSS) for 5G standalone (SA) systems, referred to as phase 1 and phase 2 of DSS respectively, are potential solutions to address the anticipated surge in spectrum demand in the coming years.

This paper investigates several network enhancement techniques applicable in the present or near future. For such purpose, a comprehensive methodology for constructing digital

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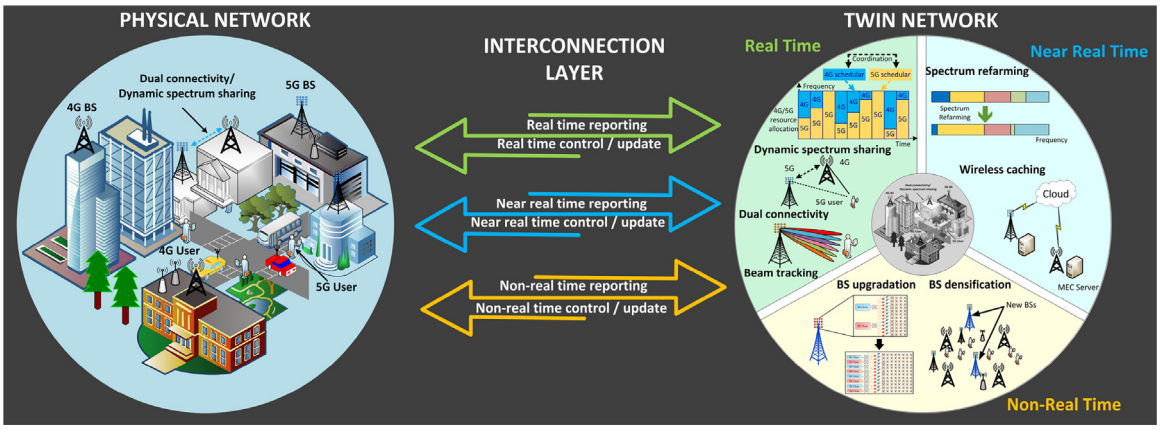

Fig. 1. Digital twin technologies for 4G, 5G, and beyond.

twin (DT) networks is introduced, enabling an accurate emulation of physical 4G/5G networks. The main contributions of this paper are as follows.

- A novel framework for constructing DT networks for 4G and 5G cellular systems, which uses measurement data from UEs and geographical information and characteristics of 4G and 5G BSs, has been developed.
- Two mapping methods, based on curve fitting and cumulative distribution function (CDF), have been proposed to analyze the relationship for field measurement between signal-to-interference-plus-noise ratio (SINR) and SE, which play a key role in evaluating 4G and 5G network performance and management techniques.
- Case studies have been provided incorporating the field measurement data and system-level simulations to demonstrate the usefulness of DT enabled network management. In particular, the impacts of DC and DSS in 4G and 5G networks have been evaluated as a real time DT application and long-term network improvement strategies such as BS densification, advanced multiple input and multiple output (MIMO) technologies, and additional spectrum allocation have been evaluated as a non-real time DT application.

2. DT enabled network management

Because of a sharp rise in the price of legacy spectrum bands as explained previously, efficient operation of expensive 4G and 5G spectrum bands is crucially important [8,9]. It is worthwhile to mention that DT technologies can be used for managing existing networks. In [10], a cloud-based concept of utilizing DTs has been introduced for the continuous assessment and maintenance of 5G and beyond networks. In [11], the potential role of DTs and modular DTs for the 6G network design, development, and operation has been discussed. Also, fault diagnosis of 5G networks has been considered based on a DT model in [12].

As illustrated in Fig. 1, depending on the update period or data gathering and action execution cycle, we might divide DT enabled network management techniques into three categories: real time DTs, near real time DTs, and non-real time DTs. Note that traditional 4G and 5G system-level simulators have primarily focused on general network performance evaluation, reflecting the statistical characteristics of representative environments such as urban or suburban areas [8,9]. In contrast, the proposed DT enabled network management aims to manage or predict a specific real-world network area by constructing its DT network. Specifically, our target area encompasses approximately 2.3 km^2 region near Gangnam subway station in South Korea as illustrated in Fig. 2, which is the most crowded and congested economic hub in the country.

2.1. Field measurements for digital twin networks

We collected field measurements for the DT networks with one of the three major mobile network operators: Korea Telecom (KT), SK Telecom (SKT), and LGU+ for reference. The data was gathered in July 2022 during peak hours of the day. Three types of traffic were independently tested: File transfer protocol (FTP) file downloads, video on demand (VoD) with YouTube videos of 720p quality, and NaverBlog webpage downloads. From the measurement devices, diagnostic monitoring and logging (DML) and diagnostic monitoring analyzer (DMA) as shown in Fig. 2, we extracted two kinds of data sets from the collected wireless signals in a slot level (1 ms for 4G slot and 0.5 ms for 5G slot): The first is *channel quality data*, which is measured by the DML through observing wireless signals and it records the SINR, reference signal received power (RSRP), reference signal received quality (RSRQ), and reference signal strength indicator (RSSI) values over time. The second is *system control data* which is obtained by decoding the physical downlink control channel (PDCCH) and contains information such as modulation and coding scheme (MCS) index, modulation order, transport block (TB) size, resource block (RB) number, etc. Note that the system control data can be utilized to calculate the instantaneous SE at each slot.

3. SE estimation via measurement data

The SE is the most fundamental and useful metric when evaluating system performance. While SE can be generally

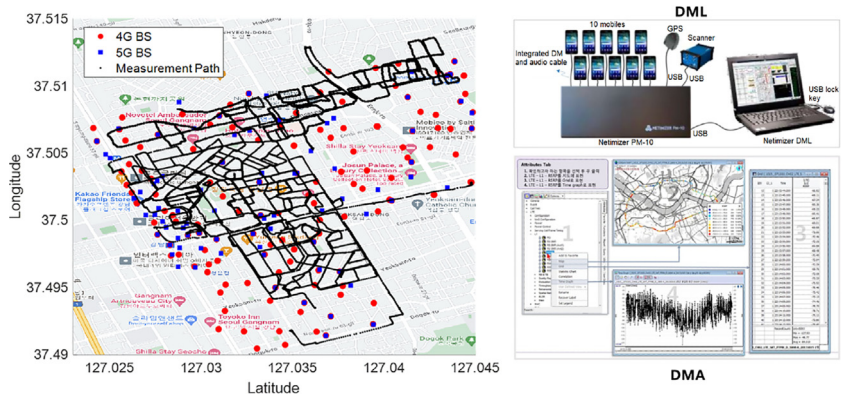

Fig. 2. Measurement area and measurement devices.

obtained through link-level simulations (LLSs), it cannot accurately reflect SE in practical systems. In addition, for performing LLS, we need to carefully consider the relations among the SINR, channel quality indicator (CQI), MCS index, TB index, RB number, etc. in 4G/5G systems. Alternatively, in this section, we introduce a methodology for obtaining SE over SINR based on field measurement.

Firstly, to calculate SE, we need to obtain the TB size R_{TB} (bytes) and the corresponding RB number N_{RB} from the system control data. Denote t_0 and B_0 respectively as the duration of a subframe for 4G (or a slot for 5G) and the bandwidth of each RB. Then we have $SE = \frac{8\times R_{TB}}{N_{RB}\times t_0\times B_0}$(bits/s/Hz).

Secondly, we need to map SE to the corresponding SINR obtained from the channel quality data. As discussed before, one difficulty of mapping SINR–SE is that the recording times of the channel quality data and the system control data are not synchronized as they are recorded by the DML independently. Note that system control data was sent (or generated) by the BS while SINR was measured by the UE at its scheduled time. In this article, we applied two mapping methods as follows:

Curve fitting: With the (partial) data batches containing synchronized SINR and SE pairs in the measurement data, we can apply the second degree polynomial curve fitting. Based on our measurement data, we obtained the SINR–SE mapping as follows.

$$SE_{4G} = 0.00521x^2 + 0.1576x + 0.2778, \quad (1)$$

$$SE_{5G} = 0.000398x^2 + 0.468x + 3.546, \quad (2)$$

where x denotes the SINR.

CDF mapping: For the most of data batches with asynchronous recording times for SINR and SE in the measurement data, it is not possible to couple those two directly. As an alternative approach, we begin by constructing the empirical CDF of the SINR values, denoted as $F_{\text{SINR}}(x)$, as well as that of the SE, denoted as $F_{\text{SE}}(y)$. Then, by applying the fact that the mapping between SINR and SE is non-decreasing, we map the SINR and SE values showing the same CDF values. Mathematically, we can write

$$y = F_{SE}^{-1}(F_{SINR}(x)),$$
$$F_{SINR}(x) \in (0, 1), x \in D_{SINR},$$

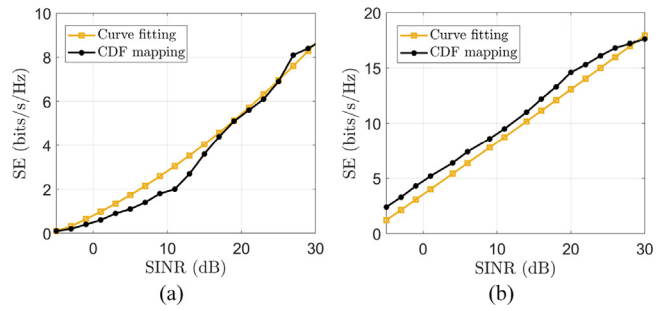

Fig. 3. SINR–SE mapping results: (a) 4G SE results and (b) 5G SE results.

$$F_{SE}(y) \in (0, 1), y \in D_{SE}, \quad (3)$$

where x and y denote the SINR and SE while D_{SINR} and D_{SE} are the corresponding ranges of these values in the measurement data. The CDF mapping effectively uses the monotonic correspondence between SE and SINR to extract a reasonable relationship from a large amount of data. As the amount of data increases, the functional relationship becomes more reliable.

It is worth highlighting that the CDF mapping can efficiently overcome the mapping problem between the two asynchronous data sets. For the field measurement, we collected wireless signals in the Gangnam area in South Korea and estimated the SINR–SE mapping for 4G and 5G with the proposed two mapping methods. Fig. 3 shows the corresponding mapping results.

4. DT enabled network management: Use cases

In Section 2, we discussed real time, near real time, and non-real time DT applications for network management and prediction. To provide insight into how these types of DTs can be applied in real-world environments incorporated with field measurement data, we present demonstrative use cases in this section. Specifically, we focus on the impact of network densification and upgrade in terms of 5G traffic management as non-real time DT use cases. We further consider DC and DSS between 4G and 5G systems to validate the benefit of

feasible spectrum utilization for network operation as near real or real time DT use cases.

The overall methodology for DT enabled system-level simulation is illustrated in Fig. 4. In the top left figure, 4G and 5G measuring equipment are used to gather empirical data for various application-level services such as FTP, hypertext transfer protocol (HTTP), video, voice over internet protocol (VoIP), etc. Data analysis and refinement based on this measurement data are performed to construct digital replicas of 4G or 5G physical networks, as depicted in the bottom left figure. Along with measurement-based mapping functions, map-based geometric information, and related parameters are utilized to construct digital-twined 4G and 5G networks [13]. Measurement-based traffic models for different applications are employed for spectrum usage estimation and prediction, as shown in the top right figure. Finally, temporal–spectral scheduling or resource allocation is performed by integrating these DT components into a system-level simulator, as depicted in the bottom right figure. Next, we present the network model and simulation scenario for the proposed system-level simulator.

4.1. Network model and simulation scenario

For simulations, we define an observation region based on real geometric data and create a comprehensive system that incorporates DT techniques. The proposed DT enabled simulation includes both 4G and 5G BSs reflecting real-world deployment and suppose that there are L 4G BSs and M 5G BSs in the observation region. Notably, we integrate advanced techniques such as DC and DSS into the simulation model, which allow for the simultaneous service of UEs by both 4G and 5G BSs. We focus on the downlink MIMO transmission in which each of 4G and 5G BSs are equipped with $N_{tx}^{[4G]}$ and $N_{tx}^{[5G]}$ antennas respectively and each of 4G and 5G UEs are equipped with $N_{rx}^{[4G]}$ and $N_{rx}^{[5G]}$ antennas respectively. Furthermore, for both the BS and UE cases, each antenna is assumed to be connected to a separate transceiver (TRX) equipment. Note that we consider a single-user MIMO support in a given RB. For the sake of clarity in exposition, we simplify the notation by disregarding the RB and UE indices. Under this assumption, the three-dimensional (3D) channel $\mathbf{H}_l^{[4G]} \in \mathbb{C}^{N_{rx}^{[4G]} \times N_{tx}^{[4G]}}$ from the lth 4G BS to its serving UE is given as $\mathbf{H}_l^{[4G]} = 10^{-\sqrt{\frac{PL+SL}{10}}} \tilde{\mathbf{H}}_l^{[4G]}$, where $\tilde{\mathbf{H}}_l^{[4G]}$ is the Rayleigh fading channel matrix, and each entry follows a complex normal distribution $\mathcal{CN}(0,1)$. Here, PL and SL denote the 3D path-loss and shadowing loss, which are functions of the distance and carrier frequency (see [5]). Then, the received signal at the serving UE is expressed as

$$\mathbf{y}_l^{[4G]} = \mathbf{H}_l^{[4G]}\mathbf{V}_l^{[4G]}\mathbf{s}_l^{[4G]} + \sum_{i=1, i\neq l}^{L} \mathbf{G}_i^{[4G]}\mathbf{V}_i^{[4G]}\mathbf{s}_i^{[4G]} + \mathbf{n}_l^{[4G]}. \quad (4)$$

Here, $\mathbf{V}_l^{[4G]} = \mathbf{H}_l^{[4G]H}\left(\mathbf{H}_l^{[4G]}\mathbf{H}_l^{[4G]H}\right)^{-1}\sqrt{\mathbf{P}_l^{[4G]}} \in \mathbb{C}^{N_{tx}^{[4G]} \times N_{rx}^{[4G]}}$ is the zero-forcing precoding matrix satisfying $\|\mathbf{V}_l^{[4G]}\|^2 \leq P^{[4G]}$ [14], where $P^{[4G]} = P_0 B_0^{[4G]}$ is the total transmit power of each 4G BS, P_0 denotes the transmit power density, and $B_0^{[4G]}$ denotes the bandwidth assigned to each RB. Moreover, $\mathbf{P}_l^{[4G]} = \mathrm{diag}\left[p_{l,1}^{[4G]}, \ldots, p_{l,N_{rx}^{[4G]}}^{[4G]}\right] \in \mathbb{C}^{N_{rx}^{[4G]} \times N_{rx}^{[4G]}}$ is the power allocation matrix, $\mathbf{G}_i^{[4G]} = \left[\left(\mathbf{g}_{i,1}^{[4G]}\right)^T, \ldots, \left(\mathbf{g}_{i,N_{rx}^{[4G]}}^{[4G]}\right)^T\right]^T \in \mathbb{C}^{N_{rx}^{[4G]} \times N_{tx}^{[4G]}}$ is the channel from the ith interfering 4G BS to the serving UE, $\mathbf{s}_l^{[4G]} \sim \mathcal{CN}(\mathbf{0}_{N_{tx}^{[4G]}}, \mathbf{I}_{N_{tx}^{[4G]}})$ is the information symbol vector, and $\mathbf{n}_l^{[4G]} \sim \mathcal{CN}(\mathbf{0}_{N_{rx}^{[4G]}}, N^{[4G]}\mathbf{I}_{N_{rx}^{[4G]}})$ is the noise vector, where $\mathbf{0}_n$ denotes the $n \times 1$ all-zero vector and $\mathbf{I}_n$ denotes the $n \times n$ identity matrix. Then, the average SINR of the serving UE can be calculated as

$$\gamma_l^{[4G]} = \frac{1}{N_{rx}^{[4G]}} \sum_{j=1}^{N_{rx}^{[4G]}} \frac{p_{l.j}^{[4G]}}{\sum_{i=1, i\neq l}^{L} I_{l,i}^{[4G]} + N^{[4G]}}, \quad (5)$$

where $p_{l.j}^{[4G]} = \frac{P^{[4G]}}{N_{rx}^{[4G]}\|\mathbf{v}_{l,j}^{[4G]}\|^2}$ and $\mathbf{v}_{l,j}^{[4G]}$ denotes the jth column of $\mathbf{V}_l^{[4G]}$. Also, $I_{l,i}^{[4G]} = \sum_{n=1}^{N_{rx}^{[4G]}} p_{i,n}^{[4G]}\|\mathbf{g}_{i,n}^{[4G]}\mathbf{v}_{i,n}^{[4G]}\|^2$ denotes the interference power received from the ith interfering 4G BS, $N^{[4G]} = B_0^{[4G]} 10^{\frac{(N_0+N_f)}{10}}$ is the noise power, N_0 is the noise power density, and N_f denotes the noise figure. In a similar manner, the 3D channel for 5G (i.e., $\mathbf{H}_l^{[5G]} \in \mathbb{C}^{N_{rx}^{[5G]} \times N_{tx}^{[5G]}}$) from the mth 5G BS to its serving UE can be defined and the corresponding average SINR $\gamma_m^{[5G]}$ can be calculated. After calculating the 4G and 5G SINRs, the corresponding 4G and 5G SEs can be obtained based on the SINR–SE mapping presented in Section 3.

To calculate the resource usage, we assume that there are $K^{[4G]}$ and $K^{[5G]}$ UEs served by each 4G and 5G BSs respectively. Four different types of services (FTP, video, HTTP, and VoIP) are assumed, which are randomly assigned to each of the UEs based on certain ratios. Denote the required data rates of four services as $R_{\text{req}}^{[\text{FTP}]}$, $R_{\text{req}}^{[\text{video}]}$, $R_{\text{req}}^{[\text{HTTP}]}$, and $R_{\text{req}}^{[\text{VoIP}]}$, respectively [13,15]. For ease of simulations, we focus on RB allocation over the simulation time T, and block fading channel is assumed within such T. For the SA 4G scenario, the required data of the kth 4G UE is determined by $D_k^{[4G]} = R_{\text{req}}^{[X]}T$, where X denotes one of four services. Hence, the required service time is then calculated by $t_k^{[4G]} = \frac{D_k^{[4G]}}{SE_k^{[4G]}B_w^{[4G]}}$, where $SE_k^{[4G]}$ denotes the SE of the kth UE and $B_w^{[4G]}$ denotes the 4G system bandwidth. Consequently, the percentage of the total resource usage is computed as $\eta^{[4G]} = \frac{100}{T}\sum_{k=1}^{K^{[4G]}} t_k^{[4G]}$ %. Similarly, the resource usage for the 5G SA scenario, denoted as $\eta^{[5G]}$, can be calculated.

Unlike the SA 4G and 5G cases, a certain portion of 5G traffic can be served by the 4G bandwidth if DC or DSS techniques are assumed. In the DC case, 4G MIMO transmission operated by 4G BSs is assumed to offload 5G NSA traffic, whereas in the DSS case, 5G MIMO transmission operated by 5G SA BSs is employed while utilizing the 4G bandwidth.

4.2. Performance evaluation

The main simulation parameters are listed in Table 1. In Fig. 5, we perform simulations for DC and DSS to evaluate

Table 1
Simulation parameters.

Parameter	Value
Carrier frequency for 4G/5G	2.14 GHz/3.65 GHz
Total bandwidth $B_w^{[4G]}$/ $B_w^{[5G]}$	20 MHz/100 MHz
TX power density (P_0)	13 dBm/MHz
Noise figure (N_f)	9 dB
Noise power density (N_0)	−174 dBm/Hz
Number of TRX equipment for BS ($N_{tx}^{[4G]}/N_{tx}^{[5G]}$)	4/32
Number of TRX equipment for UE ($N_{rx}^{[4G]}/N_{rx}^{[5G]}$)	2/4
Scheduler	Round robin
Path-loss model	3GPP TR38.901
Service ratio: FTP, Video, HTTP, VoIP	0.3, 0.3, 0.2, 0.2
Required rates: $R_{req}^{[FTP]}$, $R_{req}^{[video]}$, $R_{req}^{[HTTP]}$, $R_{req}^{[VoIP]}$	9 Mbps , 4 Mbps , 2 Mbps , 0.5 Mbps
Service time (T)	1 s

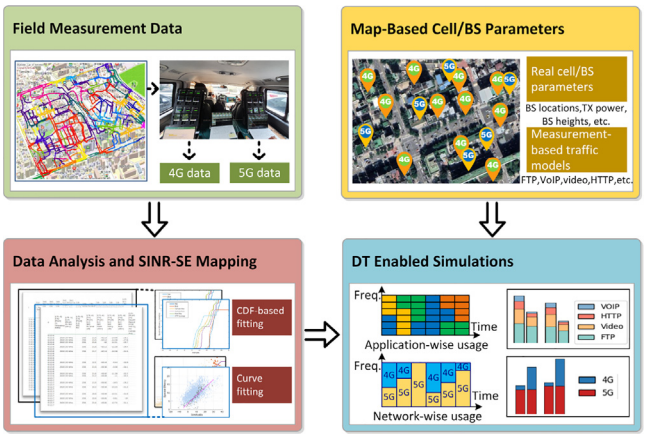

Fig. 4. DT enabled simulation methodology.

the impact of flexible spectrum usage between 4G and 5G networks and consider four types of services: video, VoIP, HTTP, and FTP. In the simulation, we focus on 5G traffic and assume that 25% of the 4G band is allowed for 5G data delivery via either DC or DSS. For DC, a part of 5G NSA traffic is delivered via 4G systems utilizing 25% of the 4G band and, therefore, a set of 4G simulation parameters in Table 1 is used. On the other hand, 5G SA systems utilize 25% of the 4G band and transmit data using 5G technologies via such 4G band. That is, a set of 5G simulation parameters in Table 1 is used for this case except for the parameters related to the carrier frequency and bandwidth. To clarify the maximum efficiency attainable by DC and DSS, we firstly schedule 5G traffic using the 4G band and then schedule the remaining traffic using the 5G band.

Fig. 5(a) plots achievable SEs for the 5G band and the 4G band utilizing DC and DSS, where the measurement-based SNR–SE mapping developed in the previous section was applied. It is seen that DSS results in improved SE than DC because the advanced 5G MIMO techniques can be used under the 4G band for the case of DSS. Fig. 5(b) plots the delivery ratio of 5G traffic via the 4G band when there exist 30 and 80 5G UEs. Because of the improved SE attained by DSS, DSS can offload approximately 260% more 5G traffic than DC for the case of 30 UEs. Figs. 5(c) and 5(d) plot the resource usages for the 5G and 4G bands, respectively. For the case of DSS, the resource usage for the 5G band decreases by approximately 43.7% than DC for the case of 30 UEs.

The performance evaluation for the non-real time DT case is given in Fig. 6, where the curve fitting-based SINR–SE mapping is used in simulation. To relieve the resource usage of those congested cells, we consider BS upgrade, additional spectrum assignment, and cell densification as seen in Figs. 6(b), 6(c), and 6(d), respectively. More specifically, the number of TRXs is upgraded from 32 to 64 and improved multi-user MIMO is introduced in Fig. 6(b), a 20 MHz bandwidth is additionally allocated in addition to the previously assigned 100 MHz bandwidth in Fig. 6(c), and additional BSs are newly deployed for the congested area in Fig. 6(d). Note that the total resource usage is reduced from an average of 53.6% to 27.2%, 45.4%, 44.1% respectively, meaning that 97.05%, 18.1%, and 21.8% more users can be accommodated respectively. Hence, upgrading BSs with a larger number of TRXs with advanced MIMO techniques can efficiently relieve the saturated spectrum usage of congested cells under a limited available spectrum and a limited number of cell sites available for equipping new BSs.

5. Conclusion

In this article, we demonstrated the potential benefits of using DTs in wireless networks, providing DT methodologies incorporated with field measurement data and geographical data. Our case studies suggested that introducing DSS for flexible spectrum usage between the 4G and 5G bands can significantly reduce the current 5G spectrum usage. Also, BS upgrade with more advanced MIMO techniques is an efficient solution for providing improved cell capacity for supporting future wireless traffic demands. However, to accommodate various real time services in the future, it is necessary to build DT networks with low latency and develop operational technologies to adjust or control communication networks in real time based on these DT networks.

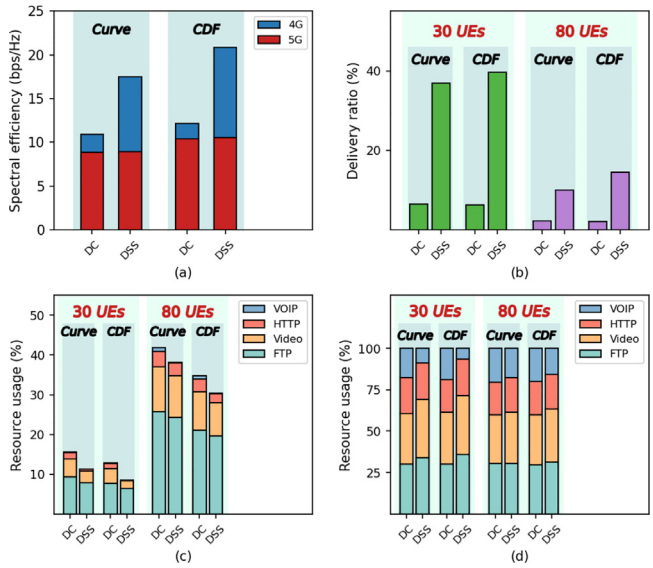

Fig. 5. Real time DT results: (a) average SEs for 4G/5G using DC and DSS; (b) delivery ratio of 5G traffic via the 4G band for DC and DSS; (c) average resource usages of the 5G band for DC and DSS; (d) average resource usages of the 4G band for DC and DSS.

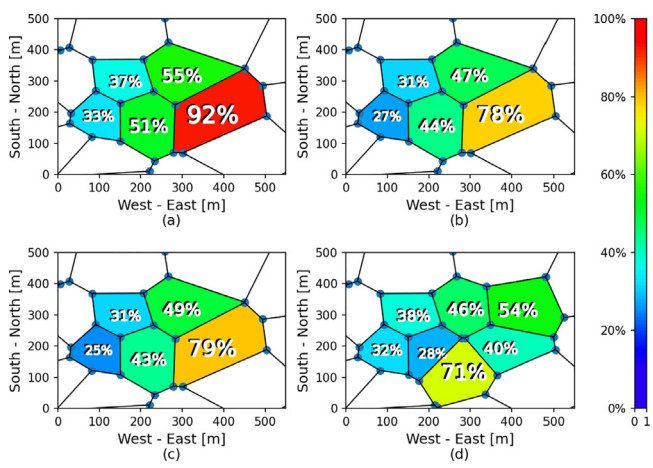

Fig. 6. Non-real time DT results: (a) average resource usage for $N_{tx}^{[5G]} = 32$ (single-user MIMO); (b) average resource usage for $N_{tx}^{[5G]} = 64$ (multi-user MIMO with 8 layers); (c) average resource usage for $B_w^{[5G]} = 120$ MHz; (d) average resource usage for two new BSs deployment.

CRediT authorship contribution statement

Najam Us Saqib: Methodology, Software, Validation, Writing. **Shilun Song:** Investigation, Methodology, Software, Validation. **Huiyang Xie:** Visualization, Data curation, Reference survey. **Zhenyu Cao:** Visualization, Data curation, Reference survey. **Gyeong-June Hahm:** Field measurements & data curation, Visualization, Simulation. **Kyung-Yul Cheon:** Field measurements & data curation, Visualization, Simulation. **Hyenyeon Kwon:** Field measurements & data curation, Visualization, Simulation. **Seungkeun Park:** Field measurements & data curation, Visualization, Simulation. **Sang-Woon Jeon:** Supervision, Conceptualization, Writing. **Hu Jin:** Supervision, Conceptualization, Writing.

Declaration of competing interest

The authors declare that there is no conflict of interest in this paper.

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